



DAFFODIL INTERNATIONAL UNIVERSITY

Department of Computing and Information System (CIS)

Discrete Mathematics

Assignment

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Assignment

Ans the Questions

Ans the Q.N-01

1(a) $\exists x(x^2=2)$

We need at least one real numbers x such that-
 $x^2=2$

Here,

$$x=\sqrt{2}, x=-\sqrt{2}$$

Both are real numbers.

Therefore, $\exists x(x^2=2) \rightarrow \boxed{\text{True}}$

1(b) $\exists x(x^2=-1)$

We need a real numbers x such that:

$$x^2=-1$$

But, $x^2 \geq 0$, So, x^2 can never be -1

So, No real number satisfies this condition.

Therefore, there is no real numbers $\exists x(x^2=-1)$
 $\rightarrow \boxed{\text{False}}$

1(c) $\forall x(x^2+2 \geq 1)$

For every real numbers x ,

$$x^2 \geq 0$$

$$\Rightarrow x^2+2 \geq 2$$

Since, $2 \geq 1$

Therefore, $x^2+2 \geq 1$ is true for every real numbers.

$\therefore \forall x(x^2+2 \geq 1) \rightarrow \boxed{\text{True}}$

1(d) $\forall x (x^2 \neq x)$

let, $x = 0 \therefore 0^2 = 0$

let, $x = 1 \therefore 1^2 = 1$

So, there exist real numbers for $x^2 = x$.

Therefore, the statement is not true for all real numbers.

$\therefore \forall x (x^2 \neq x) \rightarrow \boxed{\text{False}}$

Ans the Q.No-02

2(a)

let, $x = \text{student in this class.}$

$H(x) = x \text{ can speak hindi.}$

$\therefore$ There is a few student in this class (x)
such that x can speak Hindi
 $H(x)$

$\therefore \exists x H(x)$

If the domain is All people,

let, $x = \text{People,}$

$\therefore$ There are some people (x) such that,
 x is the student in this class and x
 $S(x)$
can speak Hindi.
 $H(x)$

$\therefore \exists x (S(x) \wedge H(x))$

2(b)

For domain = students in your class.

let, x = student in this class.

$F(x)$ = x is friendly.

$\therefore$ Every Student in this class is friendly.

$$\therefore \boxed{\forall x F(x)}$$

For Domain = All people

let, $S(x)$ = x is a student in this class.

$F(x)$ = x is friendly.

For every people(x), if x is a student in this class, then x is friendly.

$$\therefore \boxed{\forall x (S(x) \rightarrow F(x))}$$

2(c)

For domain = students in your class

let, x = student in this class.

$C(x)$ = x was born in California.

$\therefore$ There is a student in this class(x) such that x was not born in California.

$$\therefore \boxed{\exists x \neg C(x)}$$

For domain = All people

let, $S(x)$ = x is a student in this class.

$C(x)$ = x was born in California.

$\therefore$ There is a person(x) who is a student in this class and was not born in California.

$$\therefore \boxed{\exists x (S(x) \wedge \neg C(x))}$$

(d)

For Domain = students in your class,

let, x = student in this class.

$M(x)$ = x has been in a movie.

$\therefore$ There is a student in this class (x) such that x has been in a movie.

$$\therefore \boxed{\exists x M(x)}$$

For Domain = All people,

let, $S(x)$ = x is a student in this class

$M(x)$ = x has been in a movie.

$\therefore$ There is a person (x) who is a student in this class and has been in a movie.

$$\therefore \boxed{\exists x (S(x) \wedge M(x))}$$

2(e)

For Domain = students in your class,

let, x = students in this class.

$L(x)$ = x has taken a course in Logic programming.

$\therefore$ There doesn't exist any student in this class who has taken a course in logic programming.

$$\therefore \boxed{\neg \exists x L(x)}$$

For Domain = All people,

let, $S(x)$ = x is a student in this class

$L(x)$ = x has taken a course in logic programming.

For every person (x), if x is a student in this class, then x has not taken a course in logic programming.

$$\therefore \boxed{\forall x (S(x) \rightarrow \neg L(x))}$$

Ans the Q-03

3(a)

let, $P(x) = x$ is a baby.

$Q(x) = x$ is logical.

For every person (x) , if x is a baby, then x is not logical.

$$\therefore \boxed{\forall x (P(x) \rightarrow \neg Q(x))}$$

3(b)

let, $R(x) = x$ can mange a crocodile

$S(x) = x$ is despised

For every person (x) , if x can mange a crocodile, then x is not despised

$$\therefore \boxed{\forall x (R(x) \rightarrow \neg S(x))}$$

3(c)

let, $Q(x) = x$ is logical

$S(x) = x$ is despised.

for every person (x) , if x is not logical, then x is despised.

$$\therefore \boxed{\forall x (\neg Q(x) \rightarrow S(x))}$$

3(d)

let $P(x) = x$ is a baby.

$R(x) = x$ can mange a crocodile.

for every person (x) , if x is a baby then x cannot mange a crocodile.

$$\therefore \boxed{\forall x (P(x) \rightarrow \neg R(x))}$$

(Ans)

Ans the Q-04

Given:

p : Grizzly bears have been seen in the area.

q : Hiking is safe on the trail.

r : Berries are ripe along the trail.

4(a)

Berries are ripe along the trail, but grizzly have not been seen in the area.

Therefore:

$$\boxed{r \wedge \neg p}$$

4(b)

Grizzly bears have not been seen in the area and hiking on the trail is safe, but berries are ripe along the trail.

Therefore:

$$\boxed{\neg p \wedge q \wedge r}$$

4(c)

If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area.

Therefore:

$$\boxed{r \rightarrow (q \leftrightarrow \neg p)}$$

4(d)

It is not safe to hike on the trail, but grizzly bears have not been seen in the area and the berries along the trail.

$$\therefore \boxed{\neg q \wedge \neg p \wedge r}$$

4(e)

Hiking is not safe on the trail, whenever grizzly bears have been seen in the area and berries are ripe along the trail.

$$\therefore \boxed{(p \wedge r) \rightarrow \neg q} \quad (A_1)$$

Ans to the Q. No-5

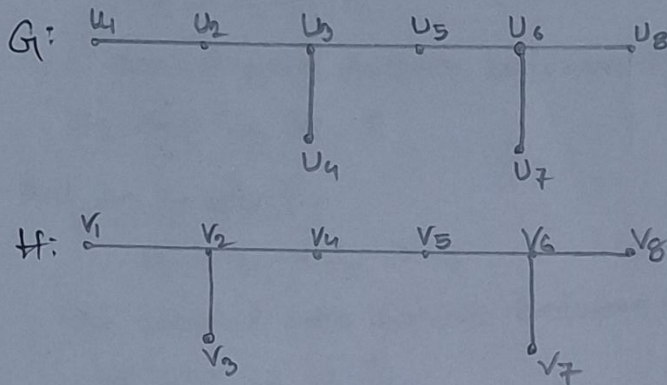


Fig: 01

Here, G_1 ,

$$\deg(u_1) = 1$$

$$\deg(u_4) = 1$$

$$\deg(u_7) = 1$$

$$\deg(u_2) = 2$$

$$\deg(u_5) = 2$$

$$\deg(u_8) = 1$$

$$\deg(u_3) = 3$$

$$\deg(u_6) = 3$$

Using Handshaking Theorem,

$$\sum \deg(u) = 2E_1$$

$$\Rightarrow 1 + 2 + 3 + 1 + 2 + 3 + 1 + 1 = 2E_1$$

$$\therefore E_1 = \frac{14}{2} = 7$$

Graph H,

$$\deg(v_1) = 1$$

$$\deg(v_6) = 3$$

$$\deg(v_2) = 3$$

$$\deg(v_7) = 1$$

$$\deg(v_3) = 1$$

$$\deg(v_8) = 1$$

$$\deg(v_4) = 2$$

$$\deg(v_5) = 2$$

Using Handshaking theorem,

$$\sum \deg(v) = 2E_2$$

$$\therefore E_2 = \frac{14}{2} = 7$$

In Graph G ,

$$U_3 \rightarrow U_5 \rightarrow U_6$$

The shortest path distance between the degree-3 vertices U_3 and U_6 is 2.

And, In graph H ,

$$V_2 \rightarrow V_4 \rightarrow V_5 \rightarrow V_6$$

The shortest path distance between the degree-3 vertices V_2 and V_6 is 3.

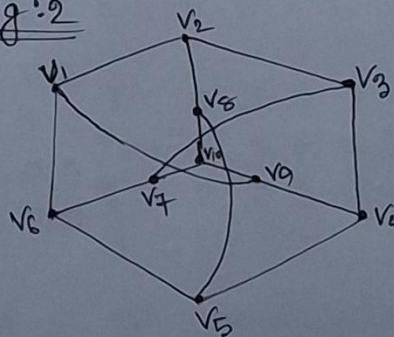
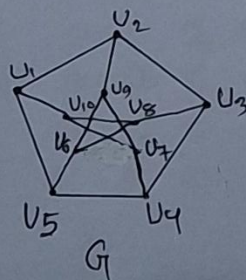
Since $G(U_3, U_6) \neq H(V_2, V_6)$ the distance between corresponding vertices is not preserved.

	U_1	U_2	U_3	U_4	U_5	U_6	U_7	U_8
U_1	0	1	0	0	0	0	0	0
U_2	1	0	1	0	0	0	0	0
U_3	0	1	0	1	1	0	0	0
U_4	0	0	1	0	0	0	0	0
U_5	0	0	1	0	0	1	0	0
U_6	0	0	0	0	0	0	1	1
U_7	0	0	0	0	0	1	0	0
U_8	0	0	0	0	0	1	0	0

	V_1	V_2	V_3	V_4	V_5	V_6	V_7	V_8
V_1	0	1	0	0	0	0	0	0
V_2	1	0	1	1	0	0	0	0
V_3	0	1	0	0	0	0	0	0
V_4	0	1	0	0	1	0	0	0
V_5	0	0	0	1	0	1	0	0
V_6	0	0	0	0	1	0	1	1
V_7	0	0	0	0	0	1	0	0
V_8	0	0	0	0	0	1	0	0

So, it is not isomorphic graph.

Fig:2



For Graph G ,

$$\begin{aligned} \deg(u_1) &= 3 & \deg(u_4) &= 3 & \deg(u_7) &= 3 & \deg(u_{10}) &= 3 \\ \deg(u_2) &= 3 & \deg(u_5) &= 3 & \deg(u_8) &= 3 \\ \deg(u_3) &= 3 & \deg(u_6) &= 3 & \deg(u_9) &= 3 \end{aligned}$$

Now, Using Handshaking theorem,

$$\sum \deg(u) = 2E_1$$

$$\Rightarrow 30 = 2E_1$$

$$\Rightarrow E_1 = 15$$

For Graph H ,

$$\begin{aligned} \deg(v_1) &= 3 & \deg(v_4) &= 3 & \deg(v_7) &= 3 & \deg(v_{10}) &= 3 \\ \deg(v_2) &= 3 & \deg(v_5) &= 3 & \deg(v_8) &= 3 \\ \deg(v_3) &= 3 & \deg(v_6) &= 3 & \deg(v_9) &= 3 \end{aligned}$$

Now, Using Handshaking thm,

$$\sum \deg(v) = 2E_2$$

$$\Rightarrow 30 = 2E_2$$

$$\Rightarrow E_2 = 15$$

Here the vertex mapping,

$$u_1 = v_1 \quad u_4 = v_7 \quad u_7 = v_3 \quad u_{10} = v_2$$

$$u_2 = v_3 \quad u_5 = v_6 \quad u_8 = v_8$$

$$u_3 = v_{10} \quad u_6 = v_5 \quad u_9 = v_4$$

	u_1	u_2	u_3	u_4	u_5	u_6	u_7	u_8	u_9	u_{10}		v_1	v_2	v_3	v_4	v_5	v_6	v_7	v_8	v_9	v_{10}
u_1	0	1	0	0	1	0	0	0	0	1	v_1	0	1	0	0	1	0	0	0	0	1
u_2	1	0	1	0	0	0	0	0	0	1	v_2	1	0	1	0	0	0	0	0	1	0
u_3	0	1	0	1	0	0	0	1	0	0	v_3	0	1	0	1	0	0	0	1	0	0
u_4	0	0	1	0	1	0	1	0	0	0	v_4	0	0	1	0	1	0	1	0	0	0
u_5	1	0	0	0	1	0	1	0	0	0	v_5	1	0	0	1	0	1	0	0	0	0
u_6	0	0	0	0	1	0	0	1	1	0	v_6	0	0	0	0	1	0	0	1	1	0
u_7	0	0	0	1	0	0	0	0	1	1	v_7	0	0	0	1	0	0	0	0	0	1
u_8	0	0	1	0	0	1	0	0	0	1	v_8	0	0	1	0	0	1	0	0	0	1
u_9	0	1	0	0	0	1	1	0	0	0	v_9	0	1	0	0	0	1	1	0	0	0
u_{10}	1	0	0	0	0	0	1	1	0	0	v_{10}	1	0	0	0	0	0	1	1	0	0

Therefore, Graph G and H are isomorphic.